

CASE 2

THE TRAVEL LIGHT DATABASE

Designing a Relational Database to Create Tables, Forms, Queries, and Reports

PREVIEW

In this case, you will design a relational database for a business that stores, packs, and ships baggage for business travelers. After your design is complete and correct, you will create database tables and populate them with data. Then you will produce one form, five queries, and two reports. The queries will address the following questions or tasks: Which customers have a credit card on file that will soon expire? Which customers travel the greatest number of days? Which customers take the most trips? What are the details of a specified trip? Which trips need to be canceled? Your reports will summarize the inventory of each customer's bag and the trip itinerary for each customer.

PREPARATION

- Before attempting this case, you should have some experience in database design and in using Microsoft Access.
- Complete any part of Tutorial A that your instructor assigns.
- Complete any part of Tutorial B that your instructor assigns, or refer to the tutorial as necessary.
- Refer to Tutorial F as necessary.

BACKGROUND

You love to travel and hope someday to have a job that allows you to do so. You probably caught the travel bug from your grandmother, Ellen, who traveled for years as a top executive for a large consulting firm. Ellen retired in her early 60s and now is beginning a new venture called Travel Light to assist business travelers. The company stores, cleans, and launders the contents of travelers' bags, then packs and ships their suitcases to specified destinations. In other words, frequent travelers can send their business attire to Travel Light. When they need their garments and other effects, Travel Light packs the bags and ships them to the assigned destinations. When travelers return from the trip, Travel Light receives the bags, dry cleans the clothes, and stores them for the next trip. Given the cost and hassle of checking bags these days, Ellen thinks the business model is strong.

At a recent family reunion, you began chatting with Ellen about your information systems course. She found out that you are proficient in designing and implementing databases, so she hired you for the summer to create a prototype system for her new business. The Access database will be a model for Travel Light's future Web site.

On your first day of work, you sit down with Ellen to learn as much as you can about the business. She describes the information she needs to keep for each customer. Aside from recording customers' full names, including their courtesy titles (Dr., Mrs., Ms., or Mr.), she needs a list of their full addresses, email addresses, telephone numbers, and credit card details. Ellen anticipates that the business will grow, and she mentions that some of her customers might eventually have the same name. You suggest that an identification number would be appropriate to uniquely identify each customer.

When customers sign up with Travel Light, they send their business garments and other effects to be stored at a warehouse. Each item is logged with a unique number, the customer's name, the category of garment, and its color. The categories include trousers, blouses, and ties, for example. When customers anticipate that they

will be traveling soon, they fill out a form that specifies their destination and travel dates. Travel Light takes care of shipping the customers' baggage to the destination in time to meet the traveling customer. In the destination city, Travel Light contracts with a local firm to deliver the bag to a specified hotel.

You and Ellen discuss how the company's Web site might work. When customers book trips, they need to specify their destination city and the beginning and ending dates of their journey. You suggest that a form would be an excellent solution for this requirement.

Your grandmother's business partner, Alex, takes care of all the financial transactions. He has asked you to develop a good way to identify customers whose credit cards are expiring soon. He can then notify them of the problem and eliminate any potential for interrupted services. You realize that this issue can be easily addressed by a query.

Ellen is curious to know who her best customers are. Because the business charges by each shipment, she would like to know which customers take the most trips. She also thinks it would be useful to know which customers have the most travel days. In the future, she may be able to market directly to customers who make repeated or extended trips to a particular destination. Perhaps she could send them a fresh set of clothing, for example. You are confident that both questions can be handled by queries.

When Travel Light employees receive a request to pack and ship business attire, they want to be able to confirm that all garments are properly packed and shipped. They want to be able to examine the database quickly and make this confirmation. You explain that it is easy to create a query that accepts input and displays the required information—in this case, the destination, dates, and items to pack.

Another request has been submitted by the customer service agent, Sheila. Customers often call and ask her to remove trips that have been canceled, so Sheila wants to be able to quickly delete a trip from the database based on the destination and travel date. You know that a query will handle this request.

Finally, Ellen requests two summary reports that will be useful for review purposes. One report should list all the items, by customer, that have been handed over to Travel Light for shipment. The other report should summarize each customer's trips. You assure Ellen that these straightforward reports will be generated by the database.

ASSIGNMENT 1: CREATING THE DATABASE DESIGN

In this assignment, you design your database tables using a word-processing program. Pay close attention to the logic and structure of the tables. Do not start developing your Access database in Assignment 2 before getting feedback from your instructor on Assignment 1. Keep in mind that you need to examine the requirements in Assignment 2 to design your fields and tables properly. It is good programming practice to look at the required outputs before beginning your design. When designing the database, observe the following guidelines:

- First, determine the tables you will need by listing the name of each table and the fields it should contain. Avoid data redundancy. Do not create a field if it can be created by a calculated field in a query.
- You will need a transaction table. Think about the business event that occurs with each customer's actions. Keep in mind that some customers travel extensively. Avoid duplicating data.
- Document your tables using the table feature of your word processor. Your tables should resemble the format shown in Figure 2-1.
- You must mark the appropriate key field(s) by entering an asterisk (*) next to the field name. Keep in mind that some tables might need a compound primary key to uniquely identify a record within a table.
- Print the database design.

Table Name	
Field Name	Data Type (text, numeric, currency, etc.)
...	...
...	...

FIGURE 2-1 Table design

NOTE

Have your design approved before beginning Assignment 2; otherwise, you may need to redo Assignment 2.

ASSIGNMENT 2: CREATING THE DATABASE, QUERIES, AND REPORTS

In this assignment, you first create database tables in Access and populate them with data. Next, you create a form, five queries, and two reports.

Assignment 2A: Creating Tables in Access

In this part of the assignment, you create your tables in Access. Use the following guidelines:

- Enter at least 20 customer records that include customers' addresses, email addresses, and credit card information. Add yourself as a customer. Consider using a fake name and address generator on the Web to eliminate typing.
- For simplicity, assume there are 10 different categories of items that could be packed in a customer's bag: trousers, skirts, blouses, shirts, dresses, shoes, belts, bags, scarves, and ties.
- Populate a table that lists the contents of each customer's bag. Each bag must contain at least one item.
- Create entries for at least 50 trips.
- Appropriately limit the size of the text fields; for example, a telephone number does not need the default length of 255 characters.
- Print all tables if your instructor requires it.

Assignment 2B: Creating Forms, Queries, and Reports

You must generate one form, five queries, and two reports, as outlined in the Background section of this case.

Form

Create a form based on your Trips table (or whatever you named the table). Save the form as Trips. Your form should resemble the one in Figure 2-2.

Trips	
Trip ID	901
Customer ID	577
Destination	New York
Start Date	10/15/2017
End Date	10/29/2017

FIGURE 2-2 Trips form

Query 1

Create a select query called Customers with Expiring Credit Cards that lists all customers whose credit cards expire by a specified date. In the output, display columns for the customer's Title, First Name, Middle Initial, Last Name, Address, City, State, and Zip. Your output should resemble that in Figure 2-3, although your data will be different.

Customers with Expiring Credit Cards							
Title	First Name	Middle Initial	Last Name	Address	City	State	Zip
Mrs.	Gladys	R	Robinson	4506 Hillside Street	Phoenix	AZ	85034
Mrs.	Ana	W	Bainbridge	3518 Glendale Avenue	Woodland Hills	CA	91303
Ms.	Dorothy	P	Westra	801 Cedarstone Drive	Tiffin	OH	44883
Ms.	Sheila	G	Garcia	2038 Conaway Street	Merom	IN	47861
Mr.	Chris	B	Austin	294 Gorby Lane	Jackson	MS	39201
Mr.	Arthur	G	Love	571 Larry Street	Milwaukee	WI	53233
Dr.	Luis	J	Simon	3080 Mesa Drive	Las Vegas	NV	89102
Mrs.	Denise	D	Chew	677 Pride Avenue	New York	NY	10005
Mr.	Nell	A	Springer	4691 Meadowview Drive	Hot Springs	VA	24445
Mr.	Eric	J	Salazar	4866 Tator Patch Road	Hickory Hills	IL	60457
Mr.	Ronald	C	Blanch	406 Roosevelt Road	Mound Valley	KS	67354
Mr.	Bruce	S	McIntyre	3799 Cimmaron Road	Santa Ana	CA	92704
Dr.	Geraldine	J	Arnett	1858 Vernon Street	California	CA	21102
Mr.	Wayne	S	Reid	814 Chandler Hollow Road	Pittsburgh	PA	15212
Mr.	Randy	H	Stevens	2088 Shobe Lane	Sterling	CO	80751
Ms.	Olive	D	Pena	16 Ridenour Street	Miami	FL	33131
Ms.	Shirley	K	Ortiz	2135 Orphan Road	Eau Claire	WI	54701
Ms.	Nicole	J	Torres	3451 Ritter Street	Huntsville	AL	35816
Mr.	Jesse	S	Polson	217 Felosa Drive	Irvine	CA	92614
Ms.	Annie	E	Alli	480 Bottom Lane	Medina	NY	14103
Mrs.	Emily	K	Smith	1938 Kelly Street	Charlotte	NC	28206
Ms.	Andrea	R	Holley	1536 Private Lane	Albany	GA	31701
Mr.	Ronald	R	Whiting	4543 Counts Lane	Hartford	CT	61036
Mr.	Steven	M	Wilson	199 Pritchard Court	Marshall	MN	56258
Mr.	Ron	S	Bliss	141 Romrog Way	Burwell	NE	68823

FIGURE 2-3 Customers with Expiring Credit Cards query

Query 2

Create a query called Top Traveler that calculates the number of days each customer has traveled and then displays columns with each customer's Title, First Name, Middle Initial, Last Name, and Days Traveled. Sort the output so that the customer who has traveled the greatest number of days is shown at the top of the list. Note the column heading change from the default setting provided by the query generator. Your output should look like that in Figure 2-4, although your data will be different.

Top Traveler				
Title	First Name	Middle Initial	Last Name	Days Traveled
Mr.	Warren	C	Manning	35
Mr.	Douglas	T	Jackson	34
Mr.	George	D	Bustamante	25
Ms.	Susan	O	Warren	17
Ms.	Jennie	S	Sheffer	17
Ms.	Maria	F	Stroman	10
Mr.	Clifford	M	Felix	10
Mr.	William	D	Colby	10
Mr.	Richard	S	Welsh	5
Mr.	Randy	H	Stevens	5

FIGURE 2-4 Top Traveler query

Query 3

Create a query similar to that of Query 2. This query, called Number of Trips, calculates the number of trips each customer has taken and is booked to take. The query should include columns for Title, First Name, Middle Initial, Last Name, and Number of Trips. Sort the output so that the customer with the greatest number of trips is shown at the top of the list. Note the column heading change from the default setting provided by the query generator. Your output should resemble the format shown in Figure 2-5, but the data will be different.

Number of Trips				
Title	First Name	Middle Initial	Last Name	Number of Trips
Mr.	Warren	C	Manning	7
Ms.	Jennie	S	Sheffer	6
Mr.	George	D	Bustamante	5
Ms.	Susan	O	Warren	4
Mr.	Douglas	T	Jackson	4
Ms.	Maria	F	Stroman	2
Mr.	Clifford	M	Felix	2
Mr.	William	D	Colby	2
Mr.	Richard	S	Welsh	1
Mr.	Randy	H	Stevens	1

FIGURE 2-5 Number of Trips query

Query 4

Create a query called Trip Details that prompts for a trip ID and then displays the packing details for the specified trip. The query should include columns for Start Date, End Date, Destination, Category Name of the packed item, and its Color. Your output should resemble the format shown in Figure 2-6, but the data will be different.

Trip Details				
Start Date	End Date	Destination	Category Name	Color
11/2/2017	11/7/2017	San Jose	blouse	green
11/2/2017	11/7/2017	San Jose	skirt	White
11/2/2017	11/7/2017	San Jose	trousers	White
11/2/2017	11/7/2017	San Jose	scarf	green

FIGURE 2-6 Trip Details query

Query 5

Create a query that deletes a trip record based on a prompt for a destination city and starting date. Test the query by running it. Save your query as Cancel a Trip.

Report 1

Create a report named Bag Inventory that summarizes the contents of each customer's bag. The report's output should display headings for Last Name, First Name, Category Name, and Color. You need to create a query first to bring the fields from different tables together. Group the report on the customer's Last Name. Adjust your output so that the Last Name and First Name columns are on the same line and all fields are formatted and visible. Depending on your data, the output should resemble that in Figure 2-7. Note that only a portion of the report appears in the figure.

Bag Inventory			
Last Name	First Name	Category Name	Color
Ali	Annie	trousers	red
		dress	green
		skirt	pink
		belt	pink
Arnett	Geraldine	blouse	green
		scarf	green
		trousers	White
		skirt	White

FIGURE 2-7 Bag Inventory report

Report 2

Create a report named Trip Itinerary that summarizes each customer's trips. The report's output should show headings for the customer's full name, destination, start date of the trip, and end date. You need to create a query first to bring the fields from different tables together. Group the report on the customer's Last Name. Adjust your output so that the name columns are on the same line and all fields are formatted and visible. Depending on your data, the output should resemble that in Figure 2-8. Note that only a portion of the report appears in the figure.

Last Name	Title	First Name	Middle Initial	Destination	Start Date	End Date
All	Ms.	Annie	E	Richmond	12/16/2017	12/21/2017
Arnett	Dr.	Geraldine	J	San Jose	11/2/2017	11/7/2017
Burrows	Ms.	Martina	R	Arlington	1/21/2018	1/26/2018
Bustamante	Mr.	George	D	Pittsburgh	2/12/2018	2/17/2018
				St. Louis	2/6/2018	2/11/2018
				Salt Lake City	1/29/2018	2/3/2018
				Orlando	12/28/2017	1/2/2018
				Las Vegas	12/14/2017	12/19/2017

FIGURE 2-8 Trip Itinerary report

ASSIGNMENT 3: ANALYZING THE DATA AND MAKING A PRESENTATION

Data analytics evaluates data in order to help people make interpretations about it. A Microsoft Access database is an excellent tool for data analytics. Databases are easily scalable—that is, data can easily be added to a database, and all generated queries, forms, and reports will display the additional data, unlike a spreadsheet. For example, if the customer table includes new records, they will be displayed in any existing query that uses customer data.

The head of marketing at Travel Light, Marcus, is interested in growing the business. He would like to see a report that details the top customers—the ones who travel the most days and take the most trips. With this information, he can tailor his marketing campaigns. Your second and third queries answer these questions. Write a brief memo to Marcus explaining your results; include details from your queries' outputs. For details on writing a memo, see Tutorial E.

Create a presentation that you will use to explain the database to your grandmother and any clerical staff she might hire. Include the design of your database tables and instructions for using the database. Discuss future improvements to the database, such as customers' ability to edit the database in order to add contents to their bag. Your presentation should take less than 10 minutes, including a brief question-and-answer period.

DELIVERABLES

Assemble the following deliverables for your instructor, either electronically or in printed form:

1. Word-processed design of tables
2. Tables created in Access
3. Form: Trips
4. Query 1: Customers with Expiring Credit Cards
5. Query 2: Top Traveler
6. Query 3: Number of Trips
7. Query 4: Trip Details
8. Query 5: Cancel a Trip
9. Query for Report 1

10. Report 1: Bag Inventory
11. Query for Report 2
12. Report 2: Trip Itinerary
13. Memo
14. Presentation materials
15. Any other required printouts or electronic media

Staple all the pages together. Write your name and class number at the top of each page. Make sure that your electronic media are labeled, if required.